Dear Award Committee,

I would like to strongly and enthusiastically recommend Daisy Mugodzeri for the Distinguished Teaching Assistant Award. I have known Daisy for several years in a professional capacity as her instructor, supervisor, and mentor and I can confidently attest to her exceptional work ethic, intellectual curiosity, reliability, and strength of character. During her time as a Teaching Assistant for Introduction to Transportation Engineering (CEE 180), a large 73-student course at UCLA, Daisy played a critical role in supporting both instructional delivery and student success.

From the outset, Daisy quickly established herself as an indispensable member of the teaching team. She independently managed a substantial portion of the course's instructional and administrative responsibilities. This included organizing and uploading all course materials to BruinLearn in a clear, accessible, and well-structured format; scanning and digitizing my handwritten lecture notes weekly; and creating polished slides for formulas, equations, tables, and handouts to improve clarity and flow for students encountering dense introductory material for the first time.

Daisy also collaborated closely with me on the development of weekly assignments, discussion materials, and solution sets. She drafted discussion slides and assessment questions, sought feedback, and revised them thoughtfully before implementation. Grading was handled efficiently and consistently, with timely feedback and posted answer keys, ensuring students understood both expectations and areas for improvement.

Beyond formal responsibilities, Daisy demonstrated outstanding commitment to student support. She never missed office hours, was always punctual and prepared, and regularly went beyond scheduled times to assist students with additional questions and problem-solving. Students responded very positively to her explanations, examples during discussion sections, and the clear instructional materials she created. She was consistently accessible, responsive, and kind qualities that fostered an inclusive and welcoming learning environment.

Daisy also showed strong independence and leadership. When I was away, she independently prepared and delivered a full lecture, creating original slides aligned with the course objectives and assigned content. She coordinated closely with me to manage midterm and assignment conflicts, set multiple submission windows in BruinLearn, and proctored exams for students requiring alternative times. She attended every lecture to remain fully aligned with course content and student needs and proactively shared announcements and updates when necessary. Additionally, Daisy supported broader course engagement efforts, including organizing early-quarter "meet and greet" activities and assisting with logistics and communication for guest speakers from industry and public agencies, including senior transportation officials.

Having managed large teams as a CEO and now teaching at both UCLA and USC, I can recognize exceptional talent when I see it. Daisy is a star, an outstanding TA and an impressive

young professional who consistently aims high. What sets Daisy apart, however, is her deep sense of purpose and commitment to making a positive impact.

I give her my strongest possible endorsement for this teaching award.

Please feel free to contact me if you require further information.

Sincerely,

Professor Hasan Ikhrata

Sole Price school of Public Policy and Engineering school

University of Southern California

University of California, Los Angeles

ikhratahasan@gmail.com/hasanikh@usc.edu/hasanikh@ucla.edu

Cell (619)990-2783

Daisy Mugodzeri

Distinguished Teaching Assistant Award – Self Statement

Teaching is not a static transfer of information, but an active process of translating complex ideas into knowledge students can confidently apply in the real world. As a PhD candidate in Civil and Environmental Engineering focusing on transportation systems, my responsibility as a Teaching Assistant (TA) extends beyond helping students master technical skills. I aim to support students' intellectual confidence, curiosity, critical thinking, and sense of purpose as future engineers. In applying for the Distinguished Teaching Assistant Award, I highlight a philosophy grounded in inclusive, applied, and socially responsible learning that prepares students to address real societal challenges.

Educational excellence and innovation in Transportation engineering presents significant barriers because it combines abstract mathematical modeling with complex human behavior. Many students struggle to see how equations for traffic flow, mode choice, or roadway design connect to daily experiences like commuting, congestion, or safety. To address this, I design learning activities that ground theory in real-world contexts. In courses including Transportation Systems Analysis, Traffic Operations, and Intelligent Transportation Systems, I move beyond textbook problem-solving by incorporating active, data-driven learning. For example, in traffic flow theory, I integrated real traffic count data collected by students in Westwood, including parking behavior, roadway design, and hourly vehicle counts (**Artifact 3)**. In Intelligent Transportation Systems, students analyzed California highway data to evaluate infrastructure investments using Net Present Value and Return On Investment, assessing upgrades such as cameras, lane markings, pavements, HOV lanes, and tolling while accounting for maintenance and long-term costs. I provided hands-on feedback on weekly mini assignments, which students incorporated into final reports and presentations (**Artifact 1)**. Early correction of misconceptions allowed students to improve rapidly from the first deliverable onward, producing more accurate and cohesive final work. In Introduction to Transportation Engineering, I helped students visualize horizontal and vertical curves using familiar streets near campus, such as Westwood, to illustrate crest (u-shaped) and sag (n-shaped) curves components in geometric design of highways (**Artifact 3)**. This local framing enabled confident application of theory in assignments and exams. Inclusivity is central to my approach; I design multi-entry assignments, promote collaboration, and use clear rubrics with iterative feedback (**Artifact 2 and Artifact 1** rubrics), reducing anxiety around coding and quantitative analysis. Students consistently report increased confidence with Python, Excel, and simulation software, as well as greater motivation to pursue transportation engineering (**Artifact 1 and Artifact 3** visualizations and presentations).

My impact as a TA extends beyond classroom. I hold structured and flexible office hours for individualized support, often evolving into mentorship about research, internships, and graduate study. I connected a student interested in research with my advisor, leading to their integration into our lab, where I continue mentoring them in research. I have also proactively supported students navigating personal and transitional challenges, including family hardship, illness,

bereavement, relocation, and new employment. Through consistent check-ins, connection to campus resources, and collaboration with instructors, I helped these students regain momentum and achieve strong grades. Beyond individual mentorship, I contribute to course and departmental improvement by providing faculty feedback on assignments, exams, and lecture materials, identifying recurring challenges and misconceptions. I helped refine grading rubrics for clarity, fairness, and alignment with learning objectives, ensuring assignments accurately reflected course goals (**Artifact 2**). I approach teaching as a continuously evolving practice, reflecting on student evaluations and informal feedback to improve pacing, clarity, and accessibility (**Artifact 5**). When students struggled to bridge theory and application, I introduced scaffolded examples, real-world datasets, and trivia aligned with weekly topics to boost engagement. For instance, in discussions on roadway design standards, I highlighted using trivia how one U.S. state (Arizona) uniquely uses metric units, sparking conversations about standardization, policy, and design implications (**Artifact 3**). Beyond mandatory TA training, I pursue professional development through campus teaching workshops and independent study of inclusive pedagogy, active learning, and data-driven instruction. I have also reviewed instructional materials from Coursera, Udemy, and AWS training modules to refine content delivery for diverse students. My professional experience with learning management systems in transportation settings further strengthened my understanding of lecture organization, instructional clarity, and student-facing course design.

My philosophy centers on preparing students not merely to pass a course, but to become thoughtful, ethical, and adaptable professionals. By connecting transportation theory to public safety, infrastructure resilience, and social equity and grounding learning in real-world examples and applied projects I help students recognize the broader societal impact of their work. Teaching is central to my academic identity, and I remain committed to creating learning environments that empower students to apply engineering knowledge responsibly.

Daisy Mugodzeri

Distinguished Teaching Assistant Award – Statement of Teaching

I view transportation engineering as a social engineering discipline: decisions about signal timing, roadway design, or planning shape safety, equity, accessibility, and economic opportunity. My goal is to move students beyond rote problem-solving toward a systems-oriented mindset that integrates technical rigor with societal impact, preparing them to become adaptable, ethical professionals capable of addressing complex, societal challenges.

I view learning as an active, contextual process. In traffic flow theory, students analyze real corridor data, linking abstract equations to observable behavior. In Intelligent Transportation Systems, they evaluate roadway segments for upgrades like HOV lanes, cameras, tolling, and assess long-term feasibility through Net Present Value and Return on Investment, as illustrated in **Artifact 1**. In Transportation Systems and Policy Outcomes, students use Python and provided scripts to analyze large datasets, including the 2017 National Household Travel Survey and the California Household Travel Survey, producing visualizations that foster analytical reasoning and systems thinking, as illustrated in **Artifact 3**. I reinforce engagement with short micro-activities, such as comparing design standards across regions or global transit policies and emphasize independent and collaborative problem-solving through structured problem sets. By encouraging exploration and iteration, students develop deeper understanding, confidence, and navigate real-world trade-offs.

Inclusivity is central to my teaching. I design assignments with multiple entry points, foster collaboration, and provide transparent rubrics with constructive feedback. I also offer flexible, purposeful office hours for individualized academic and professional support. For example, I mentored two students navigating significant life transitions one balancing family responsibilities alongside coursework, and another relocating for a new job while maintaining academic commitments. Through follow-ups, transparent communication, and coordination with instructors and campus resources, both students regained momentum and completed the course. I implement blind grading practices ignoring student IDs/names, apply uniform feedback standards (**Artifact 1**), and maintain clearly communicated office hours and contact protocols.

Beyond instruction, I encourage students to pursue research, internships, and advanced coursework, linking classroom learning to long-term professional development. To reduce financial barriers, I make essential worked examples from high-cost textbooks available during discussion sections so all students can fully engage, as shown in **Artifact 4**. Support extends beyond scheduled office hours: I offer evening and weekend Zoom review sessions and remain after discussion sections for one-on-one assistance. I provide individualized feedback through BruinLearn, explaining what was incorrect, why, and how to improve, as shown in Artifacts 1 and 4. This targeted feedback enables students to strengthen weaker areas quickly and demonstrate measurable improvement across assignments. To foster engagement, I create follow-up announcements so all students including those who miss class can join early, invite students for 'meet and greet' sessions with professor, and I support community-building in lectures and

discussions by periodically providing refreshments, helping students feel welcomed and included, as shown in **Artifact 4**. I leverage multiple teaching methods to make learning active and applied. In my Intelligent Transportation Systems course, students tackle open-ended projects such as designing a managed contraflow lane to address congestion along a selected corridor of a highway road. This requires evaluating operational performance while proposing supporting infrastructure upgrades, including pavement markings, HOV lane treatments, and traffic control strategies, under realistic implementation constraints. Data-driven assignments further reinforce these skills, with weekly mini-tasks using real traffic or infrastructure datasets and formative feedback that allows students to iteratively improve before producing final reports and presentations, as illustrated in **Artifact 1** weekly assignments building into mini project and final presentation. I also incorporate "living labs" by linking classroom concepts to field observations on familiar streets for example, using Westwood Strathmore to illustrate crest (U-shaped) and sag (N-shaped) vertical curves, as illustrated in **Artifact 3** geometric design. I draw on examples from California, across the United States, and internationally to ensure that all students see diverse perspectives represented, keeping the classroom inclusive and engaging. In addition, students receive hands-on training with industry-standard simulation tools such as MATsim, VIA, and Python, Advanced Excel reinforcing theoretical concepts through realistic modeling. By the final project, students consistently report increased confidence and technical proficiency, reflected in the quality and rigor of their deliverables, as illustrated in **Artifact 3 and Artifact 1** visualizations. When analyzing projects, students consider sustainability, safety, equity, and resilience, shifting from "solving for X" to evaluating real human and societal impacts.

I approach teaching as an evolving practice, guided by formal evaluations, student feedback, and faculty mentorship, as illustrated in **Artifact 5** showing strength and weaknesses for growth. Insights from these sources inform refinements to pacing, clarity, and accessibility, and I introduce scaffolded examples, supplemental datasets, and interactive questions when students struggle to connect theory to practice. I document and share feedback on assignments and rubrics to improve course materials for future cohorts, ensuring fairness and transparency, as illustrated in **Artifact 2** 184/284 insights. Professional development strengthens my pedagogy. I completed the Mandatory TA Training (Course 495) and Professional Standards and Ethics Training (PSET), attend workshops on inclusive pedagogy and active learning, and review industry and online instructional materials to improve content delivery. I actively seek faculty feedback and facilitate collaboration through weekly meetings and communication tools like Slack, GroupMe, and Zoom, ensuring timely problem-solving and smooth course coordination. I also gained hands-on mentorship experience supervising four multidisciplinary engineering students from FAMU during the 2025 LEAP-SURE Summer Internship. The project involved empirical data analysis using Python and Excel to extract insights from the California Vehicle Choice Survey and understand household vehicle mode choices like 184/284 empirical analysis assignment, as

illustrated in **Artifact 3**. This strengthened my mentorship skills and enhanced my ability to guide students in technical analysis, interdisciplinary problem-solving, and applied learning.

My goal is to prepare thoughtful, ethical, and adaptable engineers. I emphasize technical rigor, analytical reasoning, and ethical judgment so students can evaluate transportation systems, design safe infrastructure, and make data-driven decisions that account for equity, policy, and community impact. Grounded in reflection, inclusion, and applied learning, my teaching fosters confidence, engagement, and responsibility. Through active learning and mentorship, I prepare students to translate technical knowledge into transportation solutions that serve communities sustainably and equitably.

<u>Artifact 1 - Real-World Data Assignment plus Feedback Loop (</u>*Evidence of formative assessment and rapid skill development through applied data analysis*) **Classes 286/186 284/184**

Focus: Weekly real-world data mini-assignments, Iterative, individualized formative feedback, Progressive skill development, Formative feedback informing final deliverables, uniform Rubric/Answer Key for each assignment, sample assignment questions, individual assignments build into final mini project & presentation that used prior work, detailed feedback, presentation outcome, final project report with all assignments build in

ASSIGNMENT 5

Cost Estimate for an ITS Managed Lane

Do not disclose this rubric to the students.

APPROACH

All students start with full credit (10 points) and receive deductions for the following:

- Late submittals:
  - Less than 5 minutes late: **0.25 points off.**
  - Less than one hour late: **0.5 points off.**
  - Less than one day late: **1.0 points off**
  - Less than two days late: **2.0 points off.**
  - Less than three days late: **3.0 points off.**
  - Over three days late: contact me.
- Missing files in the submittal:
  - Missing write-up of results: **2 points off.**
  - Missing Excel workbook: contact me
- Open the student's Excel workbook and perform the following checks. For any checks that do not pass, **deduct 0.25 points.**
  - Check that all green cells are filled out.
  - Check that all yellow cells are filled out.
  - Check that all blue cells are filled out.
  - Check that the value in cell A9 is $2,000,000.
  - Check that when clearing all green and yellow cells, the entire Cost Summary table goes to $0 everywhere.
  - Check that the value in R34 is ***less than*** $700,000.
- Open the "Answer Key Spreadsheet" which we will use to check some of the calculated values in the student's submittal. For any checks that do not pass, **deduct 0.25 points.**
  - First, fill in Cell A5 of the "Answer Key Spreadsheet" with the appropriate value below for each corridor shown at the end of this rubric.
  - Check that the student's values in Cells A11 and A12 match the values in the "Answer Key Spreadsheet."
  - Check that the overall project total in Cell AG73 is within 25% of the value in the "Answer Key Spreadsheet."
- Open the Write-Up of results and check for the following:
  - Missing description of the corridor at the top: **0.25 points off.**
  - Did not update the three tables or the total cost at the very end with their own numbers from their spreadsheet: **0.25 points off**.
- Finally, give the student an "answer key" for his or her reference:
  1. Copy Column AG from the Answer Key Spreadsheet and paste it into the student's submittal into Column AH, using the "Paste as Values" option.
  2. Provide the student's spreadsheet back to the student and let him or her know that you've provided a customized "Answer Key" for them in Column AH.

FEEDBACK FOR THE STUDENTS

You're probably pretty familiar with this by now; go ahead and continue to provide constructive feedback at your discretion to the students. They really do appreciate it!

ANSWER KEY SPREADSHEET

Enter this value into Cell A5 of the "Answer Key Spreadsheet" for each corridor:

- Corridor 1: **6.75**
- Corridor 2: **6.75**
- Corridor 3: **7.75**
- Corridor 4: **11.75**
- Corridor 5: **18.75**
- Corridor 6: **13.75**
- Corridor 7: **7.75**
- Corridor 8: **8.75**
- Corridor 9: **13.75**
- Corridor 10: **9.75**
- Corridor 11: **13.75**
- Corridor 12: **8.75**
- Corridor 13: **14.75**
- Corridor 14: **12.75**
- Corridor 15: **12.75**
- Corridor 16: **6.75**
- Corridor 17: **14.75**
- Corridor 18: **6.75**
- Corridor 19: **21.75**
- Corridor 20: **15.75**
- Corridor 21: **9.75**
- Corridor 22: **8.75**
- Corridor 23: **7.75**
- Corridor 24: **8.75**
- Corridor 25: **7.75**
- Corridor 26: **7.75**
- Corridor 27: **7.75**
- Corridor 28: **11.75**
- Corridor 29: **13.75**
- Corridor 30: **13.75**
- Corridor 31: **7.75**
- Corridor 32: **7.75**
- Corridor 33: **6.75**
- Corridor 34: **6.75**
- Corridor 35: **8.75**

CEE 186/286 • Spring 2025 • Course Mini-Project

Priced Managed Lane on a Congested Corridor

OBJECTIVE

Most of our transportation infrastructure in the United States is already built, and in recent decades there has been an increasing shift in priorities from designing and building facilities, to operating and maintaining them. In this mini-project, you will examine an existing congested freeway corridor with the objective of improving its operations through the application of Intelligent Transportation Systems (ITS) in the form of a new barrier-separated dynamically priced contraflow lane on the opposite side of the roadway.

DESCRIPTION

In this mini-project, you will examine the following aspects of implementing ITS:

- **Needs Assessment:** Site description and characterization of any mobility and safety deficiencies/needs on the corridor.
- **Inventory and Feasibility Assessment:** Applying a structured screening process to evaluate the feasibility of deploying the proposed ITS strategy, informed by inventory data for the existing corridor.
- **System Engineering Excerpts (from ConOps and System Requirements):** A high-level description of how the new priced managed lane will operate, along with answers to common questions from stakeholders and a short explanation for one or more system design requirements. Collectively, this information enables decision makers, executive management, stakeholders, and the general public to get a better idea of what you are proposing to do.
  - Note that a formal Concept of Operations document and System Requirements document would typically include far more detail than you will be providing here.
- **High-Level Design:** A simplified conceptual high-level diagram showing key infrastructure to give the reader a sense of how your priced managed lane will be laid out and configured in the context of your corridor.
- **Estimated Revenues and Costs:** An order-of-magnitude estimate of the revenues expected from the new priced managed lane and of the cost associated with deploying and operating the new priced managed lane, for sketch-planning purposes and setting of initial expectations.

You will also prepare a summary PowerPoint presenation and will use it to give a three-minute presentation (time limit will be strictly enforced) on the last day of class.

EVALUATION

Each element of your mini-project submittal will contribute the following amounts to your overall score:

- Needs Assessment 15%
- Inventory and Feasibility Assessment. 5%
- System Engineering Excerpts (from ConOps & System Requirements). 40%
- High-Level Design. 15%
- Estimated Revenues and Costs 15%
- Class Presentation 10%

Submitted Files: (click to load)

Assignment 2 [redacted].zip

Assessment

Grade out of 7

7

Comments for this Attempt

Great work Hao for explaining your performance metrics, and identifying differences between AM and PM peaks. VHD data align with assignment 1 reference chart showing that bottleneck starting around 12pm to 8pm marking freeflow, 7pm-we see the drop. I have attached reference charts for you to verify your VHD and TOD Plot results.

Reference TOD Charts for Corridor 17-[redacted]

MUGODZERI, DAISY, Apr 22, 2025 at 10:26

Assessment

Grade out of 7

6.75

Comments for this Attempt

Excellent work—clear assumptions, strong toll revenue estimates and net cash flow projections, and well-highlighted findings. The gross annual revenue ($3,694,370.551) for the corridor is clearly presented. Net project revenue appears incorrect and this impacted your positive cashflow year = now we are looking at past year 30. This is our stated assumption: Toll revenues are assumed to ramp up gradually, starting at 25% of long-term steady-state levels in Year 1, increasing to 50% in Year 2, and 75% in Year 3. By Year 4, due to latent and induced demand, GP lane congestion returns to original levels, and toll revenues reach 100% of steady-state levels. Your worksheet is showing wrong numbers you have Year 2 at 75% and year 3 at 90% minor mistake. This puts your Net project revenue at -$1M Year 30. I've attached the answer key for the annual revenue estimate, rounded to the nearest $0.1 million, for your reference. I also attached the correct cum cost/revenue graph with correct % assumptions.

Assignment 6 Annual Revenue - Yanling Answer Key.png
Assignment 6 Cash Flow - Yanling Correct Graph.xlsx
Yanling Correct Graph.docx

MUGODZERI, DAISY, May 19, 2025 at 10:19

lost -0.25 - Your worksheet is showing wrong numbers you have Year 2 at 75% and year 3 at 90% minor mistake.

MUGODZERI, DAISY, May 20, 2025 at 11

Download Submission Comments

Assessment

Grade out of 7

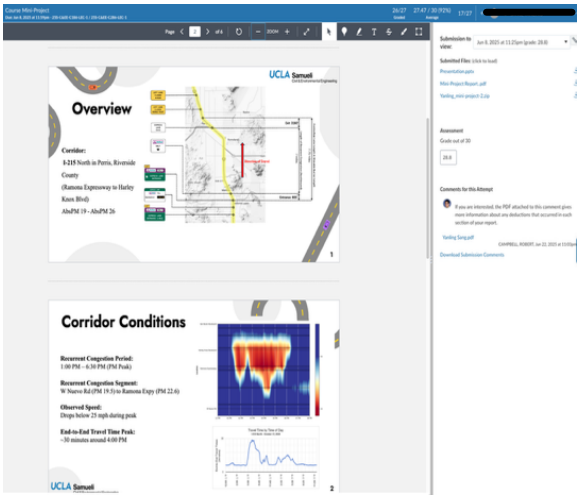
7

Comments for this Attempt

Excellent work—clear assumptions, strong toll revenue estimates and net cash flow projections, and well-highlighted findings. The net project revenue aligns well with the cumulative cost/revenue chart. The gross annual revenue ($18,401,570.05) for the corridor is clearly presented. Remember to updated the interpretation of findings to reflect your project net cumulative cash flow = positive year. is year 6. I've attached the answer key for the annual revenue estimate, rounded to the nearest $0.1 million, for your reference.

Assignment 6 Annual Revenue - Sanjay Answer Key.png

MUGODZERI, DAISY, May 19, 2025 at 12:16am



CEE 186/286 • Spring 2025

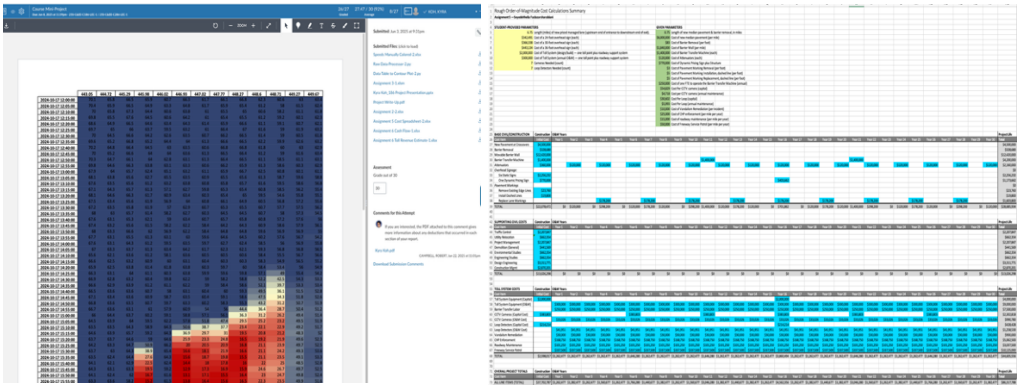
UCLA

Course Outcome

A detailed breakdown of your overall course grade is provided below. Note that specific comments and feedback are available for individual assignments via the BruinLearn gradebook, and for the course project via other pages in this PDF. If you have any questions or concerns, please feel free to reach out to me no later than 5 PM (Pacific Time) on Monday June 23. Thank you for a wonderful quarter together, and I hope you enjoyed the class!

Rising Transportation S[redacted]

GRADE CONTRIBUTION	SCORE
Assignment 1: Characterizing a Congested Corridor	7 / 7
Assignment 2: Congested Corridor Performance Metrics	7 / 7
Assignment 3: Congested Corridor Crash Analysis	7 / 7
Assignment 4: Congested Corridor ITS Inventory and Strategy Screening	7 / 7
Assignment 5: Cost Estimate for an ITS Managed Lane	7 / 7
Assignment 6: Revenue Estimate for an ITS Managed Lane	7 / 7
Removal of Lowest Assignment Score	–7 / –7
Course Mini-Project: Priced Managed Lane on a Congested Corridor	28.65 / 30
Final Exam	28 / 35
Overall Course Score	91.7 / 100
Overall Course Grade	A–



Artifact 2: Grading Rubric Refinement and Syllabus Feedback for Future Cohorts (*Evidence of instructional design that promotes clarity, fairness, and alignment with learning objectives***)**

Focus: Rubric alignment with learning objectives, Standardized lab grading for fairness and consistency, Transparent grading expectations, Syllabus and sequencing feedback for future cohorts, Faculty collaboration on instructional design

Assignments

- Assignment prompts should follow a consistent structure (overview, objectives, tasks, deliverables, grading rubric).
- Grading rubric for analytics section and project/group assignments
- For assignments with multiple components (e.g., Paper Review and Lab Analytical portions), ensure instructions clearly distinguish the requirements for each section.
- Students must follow and submit all documents listed under the Deliverables section.
    - Collaboration Policy: Students may discuss and work together, but each student must submit their own original write-up, code, and analysis. No shared files or identical answers.
- Provide writeup template/example, discussion of results, include graphs to support ideas, follow consistent order or format

Students: Common Issues to Improve

- Submit complete work.
- Answer ALL questions, including every sub-question. E.g. if it say state 2 things then mention them clearly
- Follow the order of the questions presented in the assignment. Your python code should also follow the order of the questions
- Graphs must be clearly labeled (titles, axes, legends where relevant).
- Print or show all Python output needed to verify your work.
- Use/select the appropriate graphs for your analysis (avoid irrelevant or incorrect visuals).
- Ensure visualizations are clear and properly formatted.
- Double-check that all results, code, and plots appear correctly before uploading to BruinLearn.
- If submitting code, use the correct file format: .ipynb for Jupyter notebooks, html file with printed results

Group Work

Case 1: Individual Assignments with Collaboration Allowed

- Students may work together but must produce and submit their own unique work.

Case 2: Formal Group or Team Projects

- Work collaboratively as a group, but each member submits the same agreed-upon final product to their own submission portal for record-keeping.

Submission Workflow

- Standardize required deliverables (e.g., exporting Python notebooks as HTML when requested, templates, etc).
- Apply a clear late submission penalty and communicate it consistently.

Attendance

- Clear for class, discussion, presentations

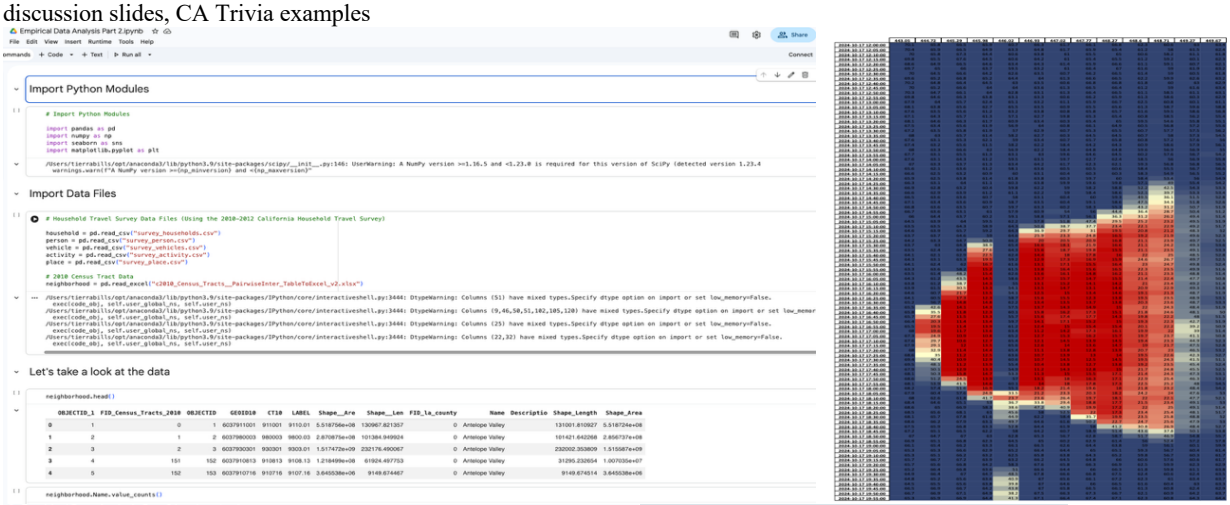
Extra items

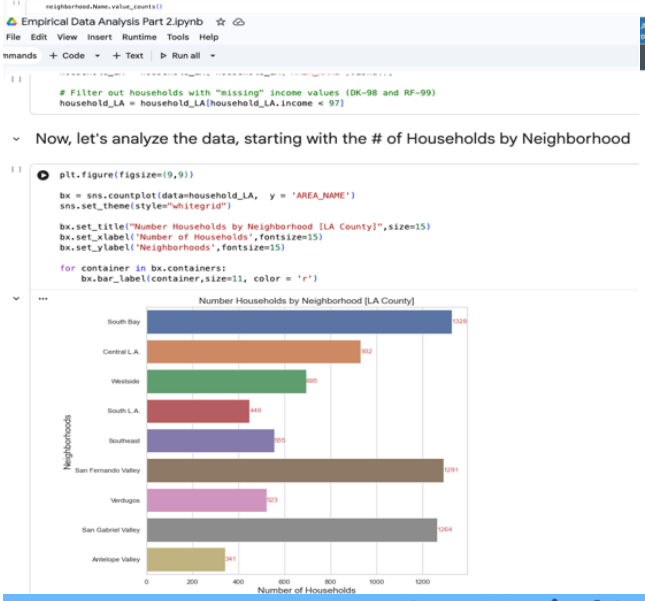
- Provide a sample completed assignment or template at the start of the quarter.
- Add a submission checklist to reduce missing components.
- Encourage use of Quarto for future reproducible reports, github
- A course website with embedded links, syllabus, schedule, etc
- Extra resources for students to access for e.g python 101, literature, case studies, etc
- Past students work as reference e.g. projects

	Category	Excellent	Good	Needs Work	Max Points
	Report Organization and Clarity/Presentation of results	(6/7 points) Report organization is clear; graphs and tables (if applicable) comumicate results effectively and are easily interpretable. All required questions/aspeccts are addressed.	(4/5 points) Report is mostly well organised, but graphs/tables are not well structured (missing labels/titles fonts too small, etc), challenging to interpret.	(<=3 points) Report is poorly-designed and organized; grammer errors; results not interpretable; does not clearly communicate findings; guidelines are largely ignored.	7/50
	Analysis Strategy is Appropriote (there will be many strategies for correctly answering the questions)	(13-15 points) All required questions have been addressed, and guidlines are followed. Approaches or algorithms that are fully appropriate for the question and data type. Demonstrates solid understanding question and how to arrive at a reasonable anwer using python script.	(6-12 points) Methods/ approach is generally appropriate but does not fully address the question. Some mismatch between method and research goal.	(<=5 points) Method is inappropriate or unjustified. Misapplied algorithms, unsupported assumptions, or no clear rationale for chosen approach.	15/50
	Correctness of Analysis Results	(9-10 points) Correct and appropriate analytical methods applied. Code runs smoothly with no errors and produces valid results. Functions and logic are clearly structured.	(6-8 points) Some errors in analysis or minor coding issues. Methods somewhat appropriate but lack justification or consistency. Code mostly runs but may need manual fixing.	(<=5 points) ncorrect or inappropriate methods used. Code contains errors or fails to run. Logic unclear or inconsistent.	10/50
	Interpretation of Results	(9-10 points) Results clearly interpreted in written responses and well-connected to assigned questions . Demonstrates critical understanding of findings.	(6-8 points) Some interpretation provided but lacks depth or misses key insights. Some results are reported without context or discussion.	(<=5 points) Little or no interpretation. Misinterpretation or no discussion of results.	10/50
	Appripriote use of data	(7-8 points) Demonstrates clear understanding of the dataset—variables, structure, limitations, and context. Selects appropriate subsets, transformations, and methods for the analysis goals. Acknowledges potential biases or uncertainties.	(5-6 points) Shows partial understanding of data content or structure. Some questionable choices in variable use or transformations. Limited awareness of data limitations.	(<=4 points) Misunderstands dataset or uses it inappropriately. Variables misapplied, context ignored, or analysis inconsistent with data limitations.	8/50
	Notes:				
	Minimim grade: Students recieve 50 points for submitting their assignment				
	If there is evidence of plagerism, a grade of 0/100 will be given				

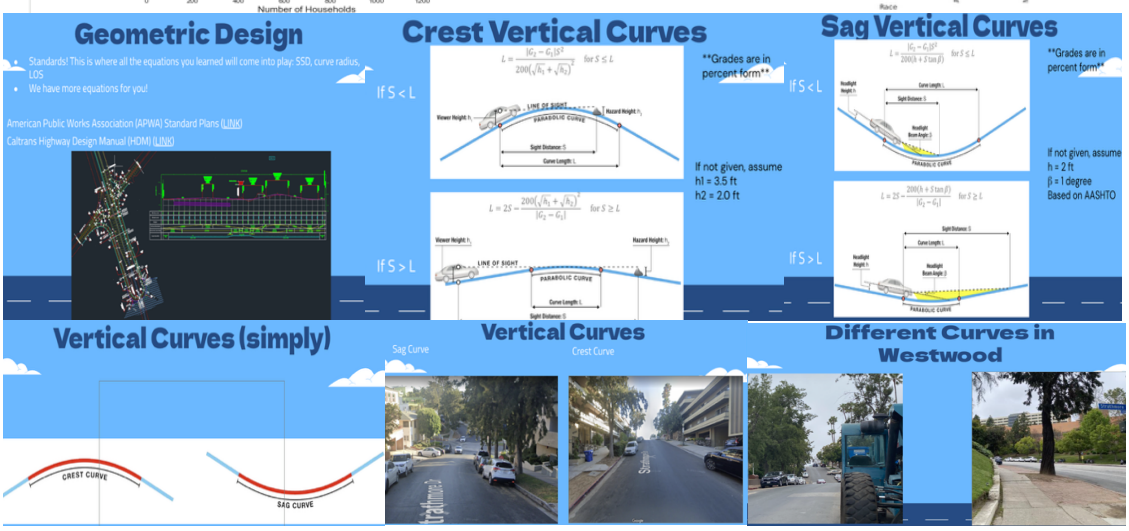
Artifact 3: Living-Lab / Visualization Teaching Material (*How students learn concepts*)

Focus: In-class visualization and applied learning, Physical or spatial examples (Westwood curves, corridor layouts), Slides or handouts used during discussion, Trivia, or visuals that trigger engagement and comprehension. 184/284 Python dataset analysis, 286/186 excel formula data analysis, 180 discussion slides geometric design, 180 use of Westwood and CA examples, 180 discussion slides, CA Trivia examples









Problem 4: (10 Points)
You are conducting a traffic study along a 300-ft section of Weastwood Boulevard, and for every 1 Hour, you note the traffic flow below for each hour:

10-minute interval	Vehicle Flow
9:00 am - 10:00 am	5
10:00 am - 11:00 am	29
11:00 am - 12:00 pm	20
12:00 pm - 1:00 pm	4
1:00 pm - 2:00 pm	14
2:00 pm - 3:00 pm	22

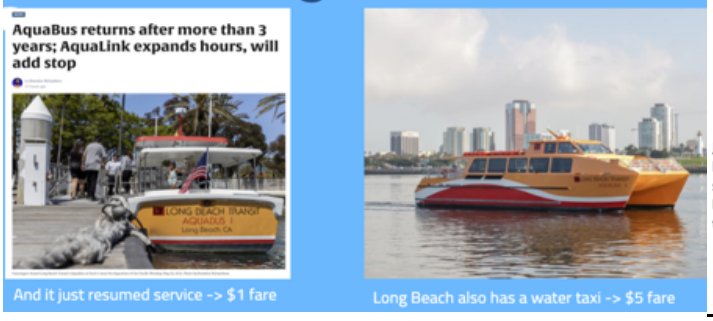
Using the collected data, what is the probability that vehicle flow in the next hour interval will be between 16-18 vehicles?

1. The four types of traffic engineering studies are: 10
   a. Traffic Volume Studies
      i. This study involves the counting of vehicles to measure traffic volume over time (i.e. how many vehicles occupy a certain area). This study could be used to determine what infrastructure (and where) needs to be improved to accommodate certain traffic patterns. For example, on a highway, having a high amount of traffic volume may suggest a new lane or exit needs to be added to help decrease bottlenecking, such as the 405 near UCLA where traffic may rack up quickly during peak hours.
   b. Speed Studies
      i. This study involves measuring the speeds of vehicles as they pass by. This study could be used to determine speed limits in certain areas as typically the speed limit is based on the 85th percentile in the speed distribution (graph of the speeds of cars in a sample). Also, it helps to determine how good speed limit enforcement is in a certain area; for example, soft speed studies could be used to determine if the cars on Gayley are driving to the speed limit, and if not, speeding deterrents can be added such as speed bumps.
   c. Travel Time and Delay Studies
      i. This study involves measuring the time it takes vehicles to travel between locations and analyzing congestion delays. This study could be used to help determine stoplight patterns; for example, in the intersection in front of Gayley Heights (Gayley and Le Conte), a substantially higher number of cars pass through Gayley than Le Conte. As a result, it would be wise to adjust the traffic light pattern by lengthening the time of the green light through Gayley to prevent congestion.
   d. Parking Studies
      i. This study involves analyzing parking demand, behavior, and occupancy rates in certain areas. This study could be used to help determine where new parking structures need to be placed and how large they should be built to accommodate for parking demand in certain areas. For example, a lot of street parking near University Apartments is taken, so performing work to create or expand on an existing parking lot structure would be great to accommodate.

Trivia

This city in Southern California has a water bus.

Long Beach, CA



Example: Poisson

John Wayne Airport in Orange County is one of the busiest single-RUNWAY airports in America. Let's say you spend an hour counting how many planes land at the airport. What is the probability that for the next 15-minute interval, 5 planes arrive?

	# Planes arriving
1-1:15	1
1:15-1:30	4
1:30-1:45	2
1:45-2	3

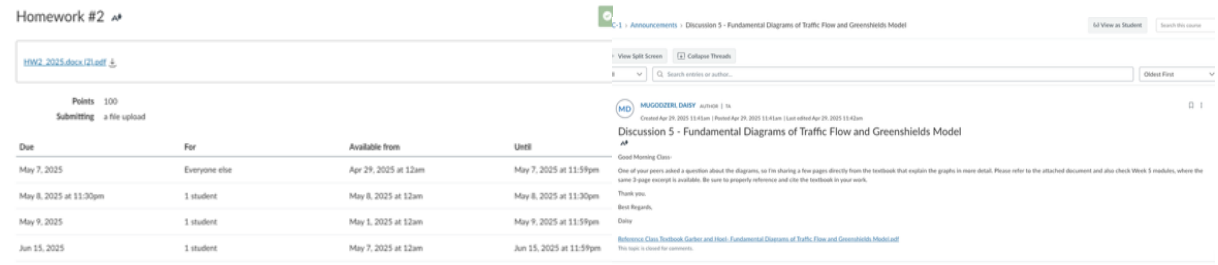
- **WHEN** - Mid Term will take place on **Monday, April 30, 2025**, during class time (**12:00-2:00 PM and you have until 2:30pm to upload all your work on bruinlearn**). Thus 30 minutes extra time to scan and upload your documents]
- **CONFLICT** - If you have any time conflicts or concerns, please contact Professor Ikhrata (and CC Ben and Daisy) as soon as possible.
- **EXAM LOCATION** - There will be a zoom link with the submission tab so you can take exam at home.
- **SUBMISSION** - All exam materials must be uploaded to Bruinlearn.
- **ACCOMMODATION** - If you need CAE accommodations we are very happy to accommodate you and will reach ou about your exam shortly.
- **SUPPORT DURING EXAM** - TAs will be available on zoom and in class during the exam time.

Re-cap 4 eras of Transportation

Reference: YouTube- Planetizen Courses' channel



Artifact 4: BruinLearn Feedback and Access Equity Evidence (*How students access and are supported in learning***)**
Focus: Supplemental resources posted to BruinLearn, Announcements ensuring equal access/submissions on different times for CAE and other students with conflicts, Anonymized written feedback, Homework Answer Keys, Textbook excerpts provided to reduce financial barriers, Documentation of inclusive instructional practices 180 announcements, 184/284 group project feedback



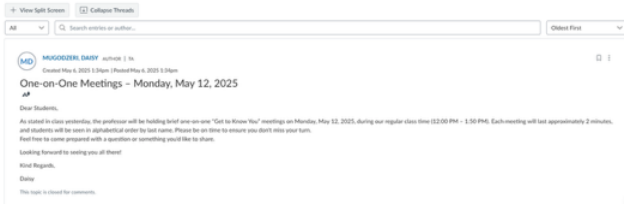
Homework #2

HW2_2025.docx (2).pdf

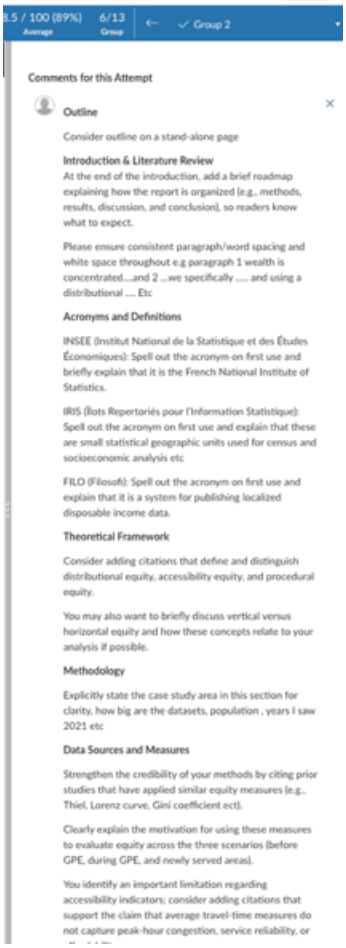
Points 100
Submitting a file upload

Due	For	Available from	Until
May 7, 2025	Everyone else	Apr 29, 2025 at 12am	May 7, 2025 at 11:59pm
May 8, 2025 at 11:30pm	1 student	May 8, 2025 at 12am	May 8, 2025 at 11:30pm
May 9, 2025	1 student	May 1, 2025 at 12am	May 9, 2025 at 11:59pm
Jun 15, 2025	1 student	May 7, 2025 at 12am	Jun 15, 2025 at 11:59pm

Discussion 5 - Fundamental Diagrams of Traffic Flow and Greenshields Model



One-on-One Meetings – Monday, May 12, 2025



8.5 / 100 (89%) Average
6/13 Group
Group 2

Comments for this Attempt

Outline

Consider outline on a stand-alone page

Introduction & Literature Review
At the end of the introduction, add a brief roadmap explaining how the report is organized (e.g., methods, results, discussion, and conclusion), so readers know what to expect.

Please ensure consistent paragraph/word spacing and white space throughout e.g paragraph 1 wealth is concentrated....and 2 ...we specifically ..... and using a distributional .... Etc

Acronyms and Definitions

INSEE (Institut National de la Statistique et des Études Économiques): Spell out the acronym on first use and briefly explain that it is the French National Institute of Statistics.

IRIS (Îlots Repertoriés pour l'Information Statistique): Spell out the acronym on first use and explain that these are small statistical geographic units used for census and socioeconomic analysis etc

FILO (Filosofi): Spell out the acronym on first use and explain that it is a system for publishing localized disposable income data.

Theoretical Framework

Consider adding citations that define and distinguish distributional equity, accessibility equity, and procedural equity.

You may also want to briefly discuss vertical versus horizontal equity and how these concepts relate to your analysis if possible.

Methodology

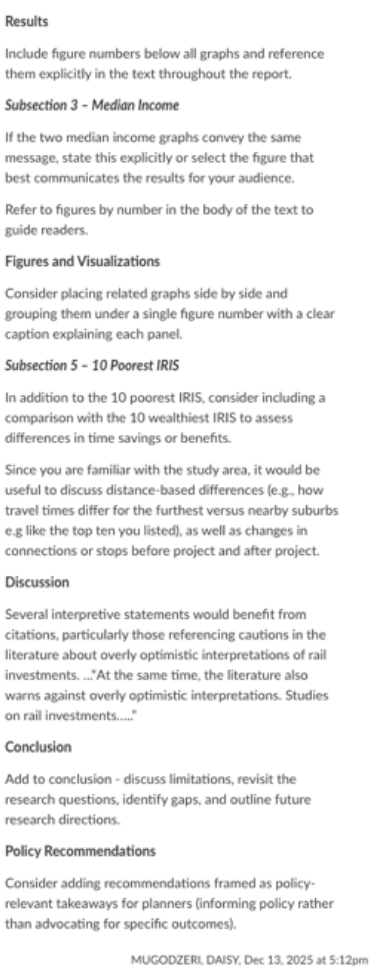
Explicitly state the case study area in this section for clarity, how big are the datasets, population , years I saw 2021 etc

Data Sources and Measures

Strengthen the credibility of your methods by citing prior studies that have applied similar equity measures (e.g., Thiel, Lorenz curve, Gini coefficient ect).

Clearly explain the motivation for using these measures to evaluate equity across the three scenarios (before GPE, during GPE, and newly served areas).

You identify an important limitation regarding accessibility indicators; consider adding citations that support the claim that average travel-time measures do not capture peak-hour congestion, service reliability, or



Results

Include figure numbers below all graphs and reference them explicitly in the text throughout the report.

Subsection 3 – Median Income

If the two median income graphs convey the same message, state this explicitly or select the figure that best communicates the results for your audience.

Refer to figures by number in the body of the text to guide readers.

Figures and Visualizations

Consider placing related graphs side by side and grouping them under a single figure number with a clear caption explaining each panel.

Subsection 5 – 10 Poorest IRIS

In addition to the 10 poorest IRIS, consider including a comparison with the 10 wealthiest IRIS to assess differences in time savings or benefits.

Since you are familiar with the study area, it would be useful to discuss distance-based differences (e.g., how travel times differ for the furthest versus nearby suburbs e.g like the top ten you listed), as well as changes in connections or stops before project and after project.

Discussion

Several interpretive statements would benefit from citations, particularly those referencing cautions in the literature about overly optimistic interpretations of rail investments. ..."At the same time, the literature also warns against overly optimistic interpretations. Studies on rail investments....."

Conclusion

Add to conclusion - discuss limitations, revisit the research questions, identify gaps, and outline future research directions.

Policy Recommendations

Consider adding recommendations framed as policy-relevant takeaways for planners (informing policy rather than advocating for specific outcomes).

MUGODZERI, DAISY, Dec 13, 2025 at 5:12pm



Indicate whether those questions were answered, identify remaining gaps, note limitations, and suggest directions for future research.

Recommendations

Since you evaluate policy in depth, consider offering recommendations. These should be framed as takeaways for planners and practitioners (informing policy rather than advocating for it).

Course-Related Measures

Consider incorporating equity measures discussed in class (e.g., Theil Index, Lorenz Curve, Gini coefficient, etc) applied to selected demographic variables or study areas.

Human Behavior Considerations

You may want to explicitly address behavioral barriers to EV adoption, drawing on existing literature (e.g., high upfront vehicle costs, charger reliability, charging costs, range anxiety etc). Are these also a contributing factors to the current discussion of adoption rates and equity?

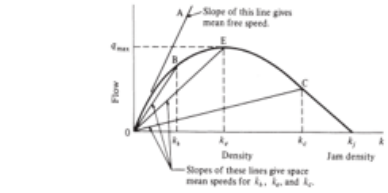
MUGODZERI, DAISY, Dec 13, 2025 at 3:09pm

6.2.1 Fundamental Diagram of Traffic Flow

The relationship between the density (veh/km) and the corresponding flow of traffic on a highway is generally referred to as the fundamental diagram of traffic flow. The following theory has been postulated with respect to the shape of the curve depicting this relationship:

1. When the density on the highway is 0, the flow is also 0 because there are no vehicles on the highway.
2. As the density increases, the flow also increases.
3. However, when the density reaches its maximum, generally referred to as the *jam density* (k_j), the flow must be 0 because vehicles will tend to line up end to end.
4. It follows that as density increases from 0, the flow will also initially increase from 0 to a maximum value. Further continuous increase in density will then result in continuous reduction of the flow, which will eventually be 0 when the density is equal to the jam density. The shape of the curve therefore takes the form in Figure 6.4a.

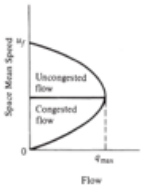
Data have been collected that tend to confirm the argument postulated above, but there is some controversy regarding the exact shape of the curve. A similar argument can be postulated for the general relationship between the space mean speed and the flow.



(a) Flow versus density



(b) Space mean speed versus density



(c) Space mean speed versus volume

Figure 6.4 Fundamental Diagrams of Traffic Flow

Equations 6.21 and 6.22 indicate that if a linear relationship in the form of Eq. 6.20 is assumed for speed and density, then parabolic relationships are obtained between flow and density and between flow and speed. The shape of the curve shown in Figure 6.4a will therefore be a parabola. Also, Eqs. 6.21 and 6.22 can be used to determine the corresponding speed and the corresponding density for maximum flow. Consider Eq. 6.21:

$$\overline{u}_s^2 = u_f \overline{u}_s - \frac{u_f}{k_j} q$$

Differentiating q with respect to $\overline{u}_s$, we obtain

$$2\overline{u}_s = u_f - \frac{u_f}{k_j}\frac{dq}{d\overline{u}_s}$$

that is,

$$\frac{dq}{d\overline{u}_s} = u_f \frac{k_j}{u_f} - 2\overline{u}_s \frac{k_j}{u_f} = k_j - 2\overline{u}_s \frac{k_j}{u_f}$$

For maximum flow,

$$\frac{dq}{d\overline{u}_s} = 0 \qquad k_j = 2\overline{u}_s \frac{k_j}{u_f} \qquad u_o = \frac{u_f}{2} \tag{6.23}$$

Thus, the space mean speed u_o at which the volume is maximum is equal to half the mean free-flow speed.

Consider Eq. 6.22:

$$q = u_f k - \frac{u_f}{k_j} k^2$$

Differentiating q with respect to k, we obtain:

$$\frac{dq}{dk} = u_f - 2k\frac{u_f}{k_j}$$

For maximum flow,

$$\frac{dq}{dk} = 0 \tag{6.24}$$

$$u_f = 2k\frac{u_f}{k_j}$$

$$\frac{k_j}{2} = k_o$$

Thus, at the maximum flow, the density k_o is half the jam density. The maximum flow for the Greenshields relationship can therefore be obtained from Eqs. 6.7, 6.23, and 6.24, as shown in Eq. 6.25:

$$q_{max} = \frac{k_j u_f}{4} \tag{6.25}$$

Table 9.20 ATS Adjustment Factor for No-Passing Zones ($f_{np,ATS}$)

Opposing Demand Flow Rate, v_o (pc/h)	*Percent No-Passing Zones*				
	≤20	40	60	80	100
		FFS ≥ 65 mi/h			
≤100	1.1	2.2	2.8	3.0	3.1
200	2.2	3.3	3.9	4.0	4.2
400	1.6	2.3	2.7	2.8	2.9
600	1.4	1.5	1.7	1.9	2.0
800	0.7	1.0	1.2	1.4	1.5
1000	0.6	0.8	1.1	1.1	1.2
1200	0.6	0.8	0.9	1.0	1.1
1400	0.6	0.7	0.9	0.9	0.9
≥1600	0.6	0.7	0.7	0.7	0.8
		FFS = 60 mi/h			
≤100	0.7	1.7	2.5	2.8	2.9
200	1.9	2.9	3.7	4.0	4.2
400	1.4	2.0	2.5	2.7	3.9
600	1.1	1.3	1.6	1.9	2.0
800	0.6	0.9	1.1	1.3	1.4
1000	0.6	0.7	0.9	1.1	1.2
1200	0.5	0.7	0.9	0.9	1.1
1400	0.5	0.6	0.8	0.8	0.9
≥1600	0.5	0.6	0.7	0.7	0.7
		FFS = 55 mi/h			
≤100	0.5	1.2	2.2	2.6	2.7
200	1.5	2.4	3.5	3.9	4.1
400	1.3	1.9	2.4	2.7	2.8
600	0.9	1.1	1.6	1.8	1.9
800	0.5	0.7	1.1	1.2	1.4
1000	0.5	0.6	0.8	0.9	1.1
1200	0.5	0.6	0.7	0.9	1.0
1400	0.5	0.6	0.7	0.7	0.9
≥1600	0.5	0.6	0.6	0.6	0.7
		FFS = 50 mi/h			
≤100	0.2	0.7	1.9	2.4	2.5
200	1.2	2.0	3.3	3.9	4.0
400	1.1	1.6	2.2	2.6	2.7
600	0.6	0.9	1.4	1.7	1.9
800	0.4	0.6	0.9	1.2	1.3
1000	0.4	0.4	0.7	0.9	1.1
1200	0.4	0.4	0.7	0.8	1.0
1400	0.4	0.4	0.6	0.7	0.8
≥1600	0.4	0.4	0.5	0.5	0.5

CEE 180 Spring 2025
Homework 2

Note: This homework is assigned on Monday, April 28, and due on Monday, May 7 (11:59 PM) uploaded electronically via BruinLearn. Please reach out if you have any questions.

Problem 1: (5 Points)
What are the four types of traffic engineering studies? Give a unique example of what each study can be used for.

a. **Spot Speed Studies . Collect data on revising speed limits.**
Example:
Study Location: Near the entrance of Fairfax High School in Los Angeles
Purpose: Evaluate driver compliance with the posted 25 mph school zone speed limit during school drop-off and pick-up hours.
Objective: Determine if additional enforcement or infrastructure (such as radar speed signs or speed humps) is needed to improve student safety.

b. **Volume Studies.**
-Setting an area over Downtown LA, count all vehicles entering or exiting the zone.
Example
Study Location: Intersection of Sunset Boulevard and La Brea Avenue
Purpose: Measure vehicle volumes during peak evening hours to determine whether a dedicated left-turn lane is justified to reduce congestion and delays.

c. **Travel Time/Delay Studies.**
-Study costs incurred when traveling along Interstate 10 during Afternoon Rush hour.
Example
Study Location: Wilshire Boulevard between Westwood and Downtown LA
Purpose: Measure average travel time and identify delay points during midday hours to evaluate the impact of bus stop locations, double parking, and signal timing on overall corridor performance.

d. **Parking Studies.**
-Count parking occupancy at a local restaurant to determine if the owner overpaid or underpaid for parking construction.
Example
Study Location: Commercial district along Melrose Avenue
Purpose: Assess occupancy rates and turnover of on-street parking during business hours to determine if time limits or dynamic pricing should be implemented to improve space availability for short-term visitors.

EQUATIONS 3.25-3.27

deceleration rate for most drivers. This rate is further justified because many studies have shown that when most drivers need to stop in an emergency, the rate of deceleration is greater than 14.8 ft/sec². Equation 3.24 then becomes

$$D_b = \frac{u^2}{30\left(\frac{a}{g} \pm G\right)} \tag{3.25}$$

Similarly, it can be shown that the horizontal distance traveled in reducing the speed of a vehicle from u to u_2 in mi/h during a braking maneuver is given by

$$D_b = \frac{u_1^2 - u_2^2}{30\left(\frac{a}{g} \pm G\right)} \tag{3.26}$$

The distance traveled by a vehicle between the time the driver observes an object in the vehicle's path and the time the vehicle actually comes to rest is longer than the braking distance, since it includes the distance traveled during perception-reaction time. This distance is referred to in this text as the *stopping sight distance* S and is given as

$$S(\text{ft}) = 1.47ut + \frac{u^2}{30\left(\frac{a}{g} \pm G\right)} \tag{3.27}$$

where the first term in Eq. 3.27 is the distance traveled during the perception-reaction time t (seconds) and u is the velocity in mi/h at which the vehicle was traveling when the brakes were applied.

EXAMPLE 3.6 – DETERMINE BRAKING DISTANCE

A student trying to test the braking ability of her car determined that she needed 18.5 ft more to stop her car when driving downhill on a road segment of 5 percent grade than when driving downhill at the same speed along another segment of 3 percent grade. Determine the speed at which the student conducted her test and the braking distance on the 5 percent grade if the student is traveling at the test speed in the uphill direction.

3.6 -SOLUTION

Solution:

- Let x = downhill braking distance on 5 percent grade.
- $(x - 18.5)$ ft = downhill braking distance on 3 percent grade.
- Use Eq. 3.25.

$$D_b = \frac{u^2}{30\left(\frac{a}{g} - G\right)}$$

$$x = \frac{u^2}{30\left(\frac{a}{g} - 0.05\right)}$$

$$x - 18.5 = \frac{u^2}{30\left(\frac{a}{g} - 0.03\right)}$$

$$18.5 = \frac{u^2}{30\left(\frac{a}{g} - 0.05\right)} - \frac{u^2}{30\left(\frac{a}{g} - 0.03\right)}$$

Determine u, the test velocity. Using $a = 11.2$ ft/sec² and $g = 32.2$,

$$\frac{a}{g} = \frac{11.2}{32.2} = 0.35$$

$$\frac{u^2}{30(0.35 - 0.05)} - \frac{u^2}{30(0.35 - 0.03)} = 18.5$$

which gives

$$0.111u^2 - 0.104u^2 = 18.5$$

from which we obtain $u = 51.4$ mi/h. The test velocity is therefore 51.4 mi/h.

- Determine braking distance on uphill grade (use Eq. 3.23).

$$D_b = \frac{51.4^2}{30(0.35 + 0.05)} = 220.16 \text{ ft}$$

Homework # 1
Due: Apr 21, 2023 at 11:59pm · 233-GAEE-180-LEC-1

Page 2 of 9 — ZOOM

Level 5: Full automation - no human intervention, the car operates independently in all conditions with no steering wheels or pedals.

Problem 4:
The transportation profession comprises many organizations represented by these acronyms. Briefly note each type of organization, its purpose, and its members.

Great work here, well done organized

20

MPO (Metropolitan Planning Organization) - Type: Regional transportation planning agency. Purpose: Responsible for regional transportation planning in urban areas, ensuring coordination between local, state, and federal agencies. Focuses on long-term transportation projects and funding allocation. Members: Local governments, transit agencies, and state DOT

representatives.

DOT (Department of Transportation) - Type: Government agency. Purpose: Oversees national or state transportation systems, including infrastructure development, safety regulations, and funding. Created in 1967 to unify transportation activities. Members: Government officials, engineers, planners, and policymakers.

FHWA (Federal Highway Administration) - Type: Federal government agency. Purpose: A sub-agency of the U.S. DOT focused on highway infrastructure, including the Interstate Highway System. Manages design, construction, and maintenance standards. Members: Highway engineers, federal and state transportation staff.

FRA (Federal Railroad Administration) - Type: Federal government agency. Purpose: Regulates and promotes safe, efficient rail transportation, including freight and passenger rail. Works under the U.S. DOT. Members: Railroad operators, safety inspectors and federal regulators.

TRB (Transportation Research Board) - Type: Research organization. Purpose: Conducts research to advance transportation innovation, policy, and best practices. Part of the National

facilities. List and briefly describe each mode's market share for passenger and freight traffic.

10

Passengers

Great work and organized

1) Highways (Personal Vehicles)
- Dominant Mode: Private automobiles account for the vast majority of passenger travel
  - 14000 - 24000 km/year per person
  - Interstate highways span 75000 km
  - Accounts for 90% of total passenger kilometers
  - Preferred for reliability, comfort, and flexibility

Good explanation

2) Aviation (Air Travel)
- Growing mode: Critical for long distance travel, nearly eliminated intercity rail.
  - 5300 public use airports
  - Post WWII expansion
  - Highest among public transit modes (about 25% of non auto passenger-km)

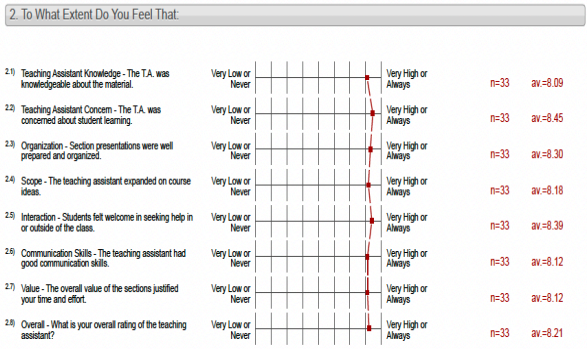
3) Rail (Passenger Rail/Transit)
- Niche mode: urban/suburban (streetcars, subways
  - Railroads peaked in early 1900s
  - Modern systems like BART serve limited metro areas.
  - 1% or less of passenger-km

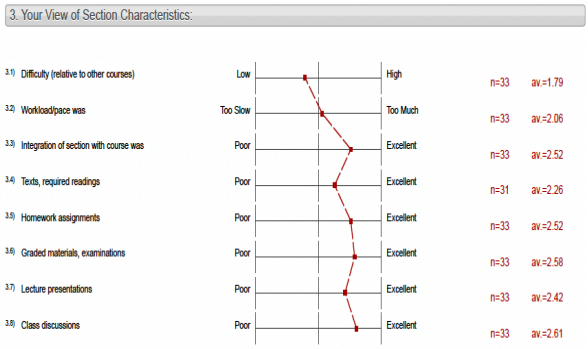
4) Public Transit
- Urban focus: Used for commuting in cities
  - Minor market share, primarily in urban and suburban areas
  - Includes fixed fixed - roue buses and paratransit

Artifact 5: Student Feedback Excerpts (SETs) (*Evidence of teaching effectiveness from the student perspective*)
Focus: Student perceptions of clarity and organization, Impact of feedback and mentorship on learning, Effectiveness of active-learning and discussion-based strategies, Growth in student confidence, engagement, and analytical skills, Qualitative evidence of real-world application of course concepts

Spring 2025 – First time being a TA at University

CEE C&EE 180 DIS 1A: INTR-TRANSPORT ENGR





Please identify what you perceive to be the real strengths and weaknesses of this teaching assistant and course.

- Although I have not been able to attend discussions for this course due to time conflicts, I believe that they were informative and helpful similar to the lectures.
- Daisy is very nice, dedicated, and knowledgeable. Discussions felt very far removed from lecture at sometimes but I think this is more of a fault of the lecturer for not covering everything in class. Paper handouts to help work on problems and reference tables would have been helpful.
- Daisy was a really nice and responsive TA. I enjoyed having her as a TA for this course. The slides prepared for discussion were very helpful, I think more emphasis on examples or discussion worksheets would be helpful with engagement!
- Daisy was an excellent TA and made her presentations very clear and easy to follow. She also posts the slides which makes it very useful when doing homework to review the content much easier and efficiently.
- Daisy was excellent at communicating and ensuring we were well prepared for our homework and exams.
- Daisy was the perfect TA. No weaknesses identified.
- Daisy was very welcoming and helpful when we asked questions.
- Discussions were extremely helpful for our homework assignments and exams. I like how the TA explained the example problems and went through them step by step.
- Fantastic and supportive TA! Very knowledgeable, easy to reach, and gave kind comments and good feedback on assignments.
- Great TA, she really cared about the students learning. Discussions were well organized.
- Great TA. Discussion always helped tremendously in understanding and applying what we learned during lecture.
- I loved this TA. I can tell she is ready to support students whenever they need it and was super responsive over email whenever students had questions. I personally feel that discussion sections can be improved to be less like students are being talked at, and maybe shortening them a bit and cutting down on the amount of information being presented in one discussion section. But other than that, I think Daisy has been super supportive and invested in our academic journey in the class and I appreciate everything she has done for her students this quarter!

- She is extremely patient and kind in my experiences with asking her for help! I felt very welcomed when I approached her with questions, and the discussion presentations that were prepared were very useful.
- Her discussions proved to be very essential for the course as they were clear and concise with relevant examples, and I do not believe I have any weaknesses I can point out.
- Strengths: Daisy is very knowledgeable on the material. She is extremely helpful outside of class. I often email my TAs for clarification because of my busy schedule and Daisy has been one to respond the quickest and also with the most detail, making sure that I learn the material and am able to apply it to the homework or discussion problems. Discussions are ran smoothly.
- Teaching assistant had good slides that showed course material well and helped me understand concepts. Some topics could be explained in a more concise and clear manner
- Teaching assistant was very welcoming and helpful with the discussion section. The discussion slides were very organized and helped fix up some of the confusion from the less organized lecture slides. The teaching assistant was also very concerned about students learning and was helpful outside of class.
- The TA Daisy is so wonderful! I really enjoyed the fun facts that were on the slides. I think the discussion slides are well organized for the most part. I only wish that we worked on the problems step by step to show how we got to the solution. The slides were definitely very helpful for the midterm and exam. Overall, I think Daisy really cared for us and did a great job!
- The TA was the best part of the course. They really cared about the lectures and relating them. Their presentations were really well put together and this is where i actually learned the material. The TA was always willing to help and make the class more interactive.
- The class was ran good. Discussion section was engaging and the practice problems were helpful for the exams and homeworks.
- Very knowledgeable TA
- she was very knowledgeable and helpful
- strengths were the amazing lectures slides presented each discussion. A weakness I'd say is the engagement for the discussion section, but overall an amazing class!!!

CEE 184 FALL 2025
C&EE 184 Transportation System and Policy Outcomes
Courses Audience: 10
Responses Received: 2
Response Ratio: 20.0%
Teaching Assistant Knowledge : The T.A. was knowledgeable about the material = Very high or Always

Teaching Assistant Concern: The T.A. was concerned about student learning = Very high or Always

Organization: Section presentations were well prepared and organized = Very high or Always

Scope: The teaching assistant expanded on course ideas = Very high or Always

Interaction: Students felt welcome in seeking help in or outside of the class = Very high or Always

Communication Skills: The teaching assistant had good communication skills = Very high or Always

Value: The overall value of the sections justified your time and effort = Very high or Always

Overall: What is your overall rating of the teaching assistant = Very high or Always

Difficulty (relative to other courses) = Medium

Workload/pace was = About Right

Integration of section with course was = Excellent

Texts, required readings = Excellent

Homework assignments = Excellent

Graded materials, examinations = Excellent

Lecture presentations = Excellent

Class discussions = Excellent

Please identify what you perceive to be the real strengths and weaknesses of this teaching assistant and course.

- Helpful

C&EE C286 DIS 1A: INT TRNSPRT SYSTEMS = No. of responses = 4 Enrollment = 10, Response Rate = 40% **186** = No. of responses = 6, Enrollment = 16, Response Rate = 37.5%

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2. To What Extent Do You Feel That:

2.1) Teaching Assistant Knowledge - The T.A. was knowledgeable about the material.	Very Low or Never	Very High or Always	n=4	av.=9.00
2.2) Teaching Assistant Concern - The T.A. was concerned about student learning.	Very Low or Never	Very High or Always	n=4	av.=9.00
2.3) Organization - Section presentations were well prepared and organized.	Very Low or Never	Very High or Always	n=4	av.=9.00
2.4) Scope - The teaching assistant expanded on course ideas.	Very Low or Never	Very High or Always	n=4	av.=9.00
2.5) Interaction - Students felt welcome in seeking help in or outside of the class.	Very Low or Never	Very High or Always	n=4	av.=9.00
2.6) Communication Skills - The teaching assistant had good communication skills.	Very Low or Never	Very High or Always	n=4	av.=9.00
2.7) Value - The overall value of the sections justified your time and effort.	Very Low or Never	Very High or Always	n=4	av.=9.00
2.8) Overall - What is your overall rating of the teaching assistant?	Very Low or Never	Very High or Always	n=4	av.=9.00

3. Your View of Section Characteristics:

3.1) Difficulty (relative to other courses)	Low	High	n=4	av.=2.00
3.2) Workload/pace was	Too Slow	Too Much	n=4	av.=2.00
3.3) Integration of section with course was	Poor	Excellent	n=4	av.=2.75
3.4) Texts, required readings	Poor	Excellent	n=4	av.=2.75
3.5) Homework assignments	Poor	Excellent	n=4	av.=2.50
3.6) Graded materials, examinations	Poor	Excellent	n=4	av.=3.00
3.7) Lecture presentations	Poor	Excellent	n=3	av.=3.00
3.8) Class discussions	Poor	Excellent	n=4	av.=3.00

4. Comments:

4.1) Please identify what you perceive to be the real strengths and weaknesses of this teaching assistant and course.

- Daisy was very patient and scrupulous when grading our assignments, and I have learnt a lot from her comments. Thanks!
- Strengths: Insightful global perspective and engaging discussions.
  Weaknesses: Pacing is a bit fast.
- The TAs are very kind and helpful!

186

2. To What Extent Do You Feel That:

2.1) Teaching Assistant Knowledge - The T.A. was knowledgeable about the material.	Very Low or Never	Very High or Always	n=6	av.=8.67
2.2) Teaching Assistant Concern - The T.A. was concerned about student learning.	Very Low or Never	Very High or Always	n=6	av.=8.50
2.3) Organization - Section presentations were well prepared and organized.	Very Low or Never	Very High or Always	n=5	av.=8.40
2.4) Scope - The teaching assistant expanded on course ideas.	Very Low or Never	Very High or Always	n=6	av.=8.33
2.5) Interaction - Students felt welcome in seeking help in or outside of the class.	Very Low or Never	Very High or Always	n=6	av.=8.50
2.6) Communication Skills - The teaching assistant had good communication skills.	Very Low or Never	Very High or Always	n=6	av.=8.50
2.7) Value - The overall value of the sections justified your time and effort.	Very Low or Never	Very High or Always	n=5	av.=8.60
2.8) Overall - What is your overall rating of the teaching assistant?	Very Low or Never	Very High or Always	n=6	av.=8.33

3. Your View of Section Characteristics:

3.1) Difficulty (relative to other courses)	Low	High	n=6	av.=2.17
3.2) Workload/pace was	Too Slow	Too Much	n=6	av.=2.00
3.3) Integration of section with course was	Poor	Excellent	n=5	av.=2.40
3.4) Texts, required readings	Poor	Excellent	n=5	av.=2.40
3.5) Homework assignments	Poor	Excellent	n=6	av.=2.83
3.6) Graded materials, examinations	Poor	Excellent	n=5	av.=2.60
3.7) Lecture presentations	Poor	Excellent	n=5	av.=2.80
3.8) Class discussions	Poor	Excellent	n=5	av.=2.60

4. Comments:

4.1) Please identify what you perceive to be the real strengths and weaknesses of this teaching assistant and course.

- Daisy was always very helpful whenever I asked her questions. She also always provided insightful feedback on the homework assignment grades.
- Daisy was an amazing TA and was always super helpful. One of the best TA's I've ever had!
- I didn't go to discussion but I appreciated daisy's responsiveness and feedback on my homework
- very helpful and responsive

Fall 2025

C&EE 184 Transportation System and Policy Outcomes

Please identify what you perceive to be the real strengths and weaknesses of this teaching assistant and course.

- Daisy was very patient and scrupulous when grading our assignments, and I have learnt a lot from her comments. Thanks!
- Strengths: Insightful global perspective and engaging discussions.
- Weaknesses: Pacing is a bit fast.
- The TAs are very kind and helpful!

Distinguished Teaching Assistant — Letter of Recommendation

This award celebrates Teaching Assistants who demonstrate a pattern of effective teaching that incorporates current best practices (e.g., student-centered learning, active learning, inclusive teaching), aims to inspire and challenge students, and supports the success of all students.

Name: Daisy Mugodzeri

Briefly describe how you know the nominee. (50 words max)

Daisy took my class in Intelligent Transportation Systems (CEE 186/286) as a graduate student in 2023 and later returned as my teaching assistant (TA) for the same class in 2025, where she provided valuable and tailored support to 26 enrolled undergraduates and graduate students.

In a few sentences, describe a few ways in which the nominee inspired or challenged you or other students in the class. (200 words max)

It's important to mention that when she took my class herself back in 2023, she did not have much background in the material and therefore had to spend additional time (more than her peers) going over extra reading, working through extra practice exercises, and attending many of my office hours for guidance/help. Her hard work and dedication paid off as she had mastered the material by the end of the quarter—but more than that, this experience proved to be invaluable to her return as a TA two years later, as she had a keen understanding and awareness of the specific difficulties, challenges, and points of confusion that many of the students themselves had, making her remarkably relatable and approachable to them. By leaning on her own experience navigating all the new and unfamiliar course content when she was in their position, she was able to explain things to the students in ways that deeply resonated with them (especially those that didn't have much pre-existing background in the subject). Furthermore, students who needed help felt comforted, reassured, and inspired knowing that she, too, had been in their position and was ultimately able to complete the class as a top performer.

Describe one or two things that the nominee did that you thought were particularly beneficial for your (or other students') learning or understanding of concepts. (200 words max)

Daisy devoted a tremendous amount of time and attention to making sure her students had all the support they needed. She organized office hours and impromptu discussion sections around their schedules, offered both online and in-person sessions to accommodate everyone's learning styles, and was happy to provide support in whichever format each student preferred (walkthroughs, examples, question-and-answer sessions, email exchange, etc.). Daisy also showed an incredible degree of investment and commitment to her students, working long hours to provide them with valuable individual feedback on their assignments, organizing guided lessons and tutorials around all the major themes/activities in the class, and responding to students' questions almost as soon as they sent them. Her students and their success were truly her top priority; though I'm sure there were many occasions throughout the quarter where her own research and classwork demanded much of her time and attention, she was never too busy for the students and always made sure to attend to their needs first. Her actions made it clear that she cared deeply about each and every one of them and their education, well-being, growth, and success.

Please share any additional reasons why you believe the nominee is an outstanding candidate for this award. (100 words max)

Daisy was not just a great instructor and tutor, she was also a great advocate and supporter too. She didn't just aim to teach them concepts and help them complete assignments—she also inspired them to persevere and thrive by modeling resilience, patience, positivity, sensitivity, gentleness, and compassion. She never missed an opportunity to be encouraging and uplifting to the students, as she provided much constructive (and individually customized) feedback and positive reinforcement to all the students on every assignment, along with tips and suggestions for advancing their skills to the next level.

Distinguished Teaching Assistant — Letter of Recommendation

This award celebrates Teaching Assistants who demonstrate a pattern of effective teaching that incorporates current best practices (e.g., student-centered learning, active learning, inclusive teaching), aims to inspire and challenge students, and supports the success of all students.

Name

Briefly describe how you know the nominee. (50 words max)

In a few sentences, describe a few ways in which the nominee inspired or challenged you or other students in the class. (200 words max)

Describe one or two things that the nominee did that you thought were particularly beneficial for your (or other students') learning or understanding of concepts. (200 words max)

Please share any additional reasons why you believe the nominee is an outstanding candidate for this award. (100 words max)

Distinguished Teaching Assistant — Letter of Recommendation

This award celebrates Teaching Assistants who demonstrate a pattern of effective teaching that incorporates current best practices (e.g., student-centered learning, active learning, inclusive teaching), aims to inspire and challenge students, and supports the success of all students.

Name Natalie Avalos

Briefly describe how you know the nominee. (50 words max)

Ms. Daisy Mugodzeri was the teaching assistant for one of the courses I was enrolled in: C&EE 180.

In a few sentences, describe a few ways in which the nominee inspired or challenged you or other students in the class. (200 words max)

Although Daisy exhibited exceptional knowledge and understanding of the course material, she inspired me the most professionally. During her office hours, she took the time to know me personally apart from academically. When I shared with her how I would have my first internship the coming summer, and how nervous I was, she shared her personal experience working her first job. This was incredibly helpful as she shared details about what to expect working in a professional engineering setting. Most of all, she reminded me that I should be confident in the skills I have developed at UCLA.

Describe one or two things that the nominee did that you thought were particularly beneficial for your (or other students') learning or understanding of concepts. (200 words max)

One of the things Daisy did that I thought was particularly beneficial to student learning was having the equations presented in class in her discussion slides. This allowed for further discussion of the concepts presented in a more relaxed setting with a smaller class size. At the beginning of each discussion section for C&EE 180 Daisy would always check in with students to make sure our doubts were resolved and would also start with a small ice breaker. In the middle of her discussion sections, she would also have a fun trivia question related to the concept at hand. Additionally, before the midterm and finals exam she also summarized key formulas we needed to know for the exam and provided ample practice problems. Because the discussion section was two hours she divided the time into lecture time and office hours. Seeing the information presented directly before office hours meant I was more well prepared figuring out the areas I was struggling in. Moreover, Daisy also recommended and included the appropriate chapters from the course textbook to help student learning.

Please share any additional reasons why you believe the nominee is an outstanding candidate for this award. (100 words max)

What makes Daisy stand out from other candidates is her kindness. In her C&EE 180 discussion sections, Daisy consistently tried to help students in any way she could. If a student needed additional office hours, she would always offer to meet up with students in person or answer their doubts through email. Moreover, as the only electrical engineering major in a civil engineering class, Daisy was extremely helpful in helping me understand concepts I wasn't necessarily familiar with. Overall, Daisy's top most priority has always been to help her students succeed.

UCLA Samueli
Civil & Environmental Engineering

Distinguished TA Award Selection Committee
University of California, Los Angeles

Dear Members of the Distinguished TA Award Committee,

I am writing to offer my strongest possible support for Ms. Daisy Mugodzeri's nomination for the Distinguished TA Award at UCLA. I support her application without any reservations. I serve as the Vice Chair for Graduate Studies in the C&EE department.

During the 2024–2025 academic year, Daisy made exceptional contributions to our teaching mission. She taught two courses in Winter 2025 (180 and 186/286) and one course in Fall (184/284). Across these courses, she consistently demonstrated outstanding preparation, clarity in her explanations, and a remarkable ability to engage students with the material at an appropriate depth and pace.

Daisy's impact is perhaps best reflected in the feedback she received from her students, which has been overwhelmingly positive. Students described her as "*super supportive and invested in our academic journey*" and remarked that "*she really cared about the students' learning*." These comments are representative of the broader sentiment in her evaluations and underscore the care and attention she devotes to her students' success.

This student feedback reflects a deep and sustained commitment Daisy has brought to her role as a teaching assistant and instructor of record. She is not only knowledgeable and organized, but she also creates an inclusive and supportive learning environment where students feel comfortable asking questions and engaging with challenging material. Her dedication to teaching, responsiveness to student needs, and genuine concern for their learning outcomes make her stand out among her peers.

In light of her exemplary teaching record, her consistently strong student evaluations, and her evident commitment to student learning, I am pleased to recommend Daisy Mugodzeri for the Distinguished TA Award without hesitation. She is exactly the kind of thoughtful, caring, and effective educator that this award is meant to recognize.

Thank you for your consideration of her nomination.

Sincerely,



Sriram Narasimhan, PhD, P.Eng (Ont), F.ASCE,

Professor and Graduate Vice Chair, C&EE